Statistical test workflows

Introduction

`testflow` provides statistical testing workflows organized by study design.

One numerical variable

```r
library(testflow)
cardio <- make_cardio_data()
test_one_sample(cardio, sbp_3m, mu = 140)
#> Statistical test workflow
#>
#> Outcome: sbp_3m
#> Design: one numerical sample
#>
#> Assumptions:
#> - Normality: acceptable
#>
#> Recommended test:
#> One-sample t-test
#>
#> Result:
#> H0: the population mean or location of sbp_3m equals the reference value.
#> statistic = -0.46, df = 179.00, p = 0.646, 95% CI [136.47, 142.20]
#>
#> Effect size:
#> Cohen's d = -0.03, negligible
#>
#> Report:
#> The one numerical sample workflow for sbp_3m did not show a statistically significant result using O
```

Two independent groups

```r
test_two_groups(sbp_3m ~ sex, data = cardio)
#> Statistical test workflow
#>
#> Outcome: sbp_3m
#> Group: sex
#> Design: two independent groups
#>
#> Assumptions:
#> - Normality by group: acceptable
#> - Homogeneity of variance: acceptable
#> - F-test variance comparison: acceptable
#>
#> Recommended test:
#> Student independent t-test
```

```
#>
#> Result:
#> H0: the population mean or location of sbp_3m is equal across levels of sex.
#> statistic = -1.91, df = 178.00, p = 0.058, 95% CI [-11.22, 0.18]
#>
#> Effect size:
#> Cohen's d = -0.29, small
#>
#> Report:
#> The two independent groups workflow for sbp_3m did not show a statistically significant result using
```

Paired measurements

```
test_paired(sbp_3m ~ sbp_baseline, data = cardio)
#> Statistical test workflow
#>
#> Outcome: sbp_3m - sbp_baseline
#> Design: paired measurements
#>
#> Assumptions:
#> - Normality of paired differences: acceptable
#>
#> Recommended test:
#> Paired t-test
#>
#> Result:
#> H0: the mean or median paired difference (sbp_3m - sbp_baseline) equals 0.
#> statistic = -9.20, df = 179.00, p = <0.001, 95% CI [-9.53, -6.16]
#>
#> Effect size:
#> Cohen's dz = -0.69, moderate
#>
#> Report:
#> The paired measurements workflow for sbp_3m - sbp_baseline showed a statistically significant result
```

More than two groups

```
test_groups(sbp_3m ~ treatment, data = cardio)
#> Statistical test workflow
#>
#> Outcome: sbp_3m
#> Group: treatment
#> Design: more than two independent groups
#>
#> Assumptions:
#> - Normality by group: not acceptable, acceptable
#> - Homogeneity of variance: acceptable
#> - Bartlett test: acceptable
#>
#> Recommended test:
#> Kruskal-Wallis test
#>
```

```
#> Result:
#> H0: the population mean or location of sbp_3m is equal across levels of treatment.
#> statistic = 7.58, df = 2.00, p = 0.023
#>
#> Effect size:
#> Kruskal epsilon squared = 0.03, small
#>
#> Report:
#> The more than two independent groups workflow for sbp_3m showed a statistically significant result u
```

Factorial designs

```
test_factorial(sbp_3m ~ sex * treatment, data = cardio)
#> Statistical test workflow
#>
#> Outcome: sbp_3m
#> Group: sex, treatment
#> Design: factorial design
#>
#> Assumptions:
#> - Residual normality: acceptable
#> - Homogeneity of variance: acceptable
#>
#> Recommended test:
#> Factorial ANOVA
#>
#> Result:
#> H0: the population mean or location of sbp_3m is equal across levels of sex, treatment.
#> statistic = 3.78, df = 1.00, p = 0.053
#>
#> Effect size:
#> eta squared = 0.02, small
#>
#> Report:
#> The factorial design workflow for sbp_3m did not show a statistically significant result using Facto
```

Repeated measurements

```
test_repeated(cardio, c(sbp_baseline, sbp_3m, sbp_6m), id = id)
#> Statistical test workflow
#>
#> Outcome: value
#> Group: time
#> Design: repeated numeric measurements
#>
#> Assumptions:
#> - Normality by time: acceptable
#>
#> Recommended test:
#> Repeated-measures ANOVA
#>
#> Result:
```

```
#> H0: the population mean or location of value is equal across levels of time.
#> statistic = 3.76, df = 2.00, p = 0.024
#>
#> Effect size:
#> eta squared = 0.05, small
#>
#> Report:
#> The repeated numeric measurements workflow for value showed a statistically significant result using
```

The repeated numeric workflow chooses repeated-measures ANOVA when the within-time normality checks are acceptable and Friedman otherwise. Post-hoc comparisons are paired t-tests for the parametric branch and paired Wilcoxon tests for the non-parametric branch.

Categorical outcomes

```r
test_categorical(treatment ~ controlled_3m, data = cardio)
#> Statistical test workflow
#>
#> Outcome: treatment
#> Group: controlled_3m
#> Design: two categorical variables
#>
#> Assumptions:
#>
#> Recommended test:
#> Chi-square test of independence
#>
#> Result:
#> H0: treatment and controlled_3m are independent.
#> statistic = 5.02, df = 2.00, p = 0.081
#>
#> Effect size:
#> Cramer's V = 0.17, small
#>
#> Report:
#> The two categorical variables workflow for treatment did not show a statistically significant result
```

Repeated categorical outcomes

```r
test_repeated_categorical(cardio, c(controlled_baseline, controlled_3m, controlled_6m))
#> Statistical test workflow
#>
#> Outcome: controlled_baseline, controlled_3m, controlled_6m
#> Design: repeated categorical measurements
#>
#> Recommended test:
#> Cochran Q test
#>
#> Result:
#> H0: the success proportions are equal across repeated categorical measures.
#> statistic = 39.58, df = 2.00, p = <0.001
#>
#> Effect size:
```

```
#> Cochran Q Kendall's W = 0.11, small
#>
#> Report:
#> The repeated categorical measurements workflow for controlled_baseline, controlled_3m, controlled_6m
```

The repeated categorical workflow uses Cochran Q for binary repeated outcomes and pairwise McNemar tests for follow-up comparisons.

References

- Fisher, R. A. (1925). *Statistical Methods for Research Workers.*
- Gosset, W. S. (1908). The probable error of a mean.
- Welch, B. L. (1947). Generalization of Student's problem with unequal variances.
- Wilcoxon, F. (1945). Individual comparisons by ranking methods.
- Mann, H. B., & Whitney, D. R. (1947). On a test of whether one of two random variables is stochastically larger than the other.
- Levene, H. (1960). Robust tests for equality of variances.
- Kruskal, W. H., & Wallis, W. A. (1952). Use of ranks in one-criterion variance analysis.
- Tukey, J. W. (1949). Comparing individual means in the analysis of variance.
- Dunn, O. J. (1964). Multiple comparisons using rank sums.
- Friedman, M. (1937). The use of ranks to avoid the assumption of normality implicit in the analysis of variance.
- Cochran, W. G. (1950). The comparison of percentages in matched samples.
- McNemar, Q. (1947). Note on the sampling error of the difference between correlated proportions or percentages.
- Pearson, K. (1895, 1900).
- Spearman, C. (1904). The proof and measurement of association between two things.
- Kendall, M. G. (1938). A new measure of rank correlation.
- Cramer, H. (1946). *Mathematical Methods of Statistics.*
- Clopper, C. J., & Pearson, E. S. (1934). The use of confidence or fiducial limits illustrated in the case of the binomial.
- Cohen, J. (1988). *Statistical Power Analysis for the Behavioral Sciences.*

Correlation

```
test_correlation(sbp_3m ~ age, data = cardio)
#> Statistical test workflow
#>
#> Outcome: sbp_3m
#> Group: age
#> Design: two numeric variables
#>
#> Assumptions:
#> - Normality: acceptable
#>
#> Recommended test:
#> spearman correlation
#>
#> Result:
#> H0: the correlation between age and sbp_3m is 0.
#> statistic = 793638.65, p = 0.014
#>
#> Effect size:
```

```
#> spearman r = 0.18, small
#>
#> Report:
#> The two numeric variables workflow for sbp_3m showed a statistically significant result using spearm
```

Outliers

```
test_outliers(cardio, c(sbp_3m, ldl, crp))
#> Statistical test workflow
#>
#> Outcome: sbp_3m, ldl, crp
#> Design: outlier screening
#>
#> Recommended test:
#> iqr outlier detection
#>
#> Result:
#> statistic = 11.00, p = NA
#>
#> Effect size:
#> Outlier count = 11.00, NA
#>
#> Report:
#> The outlier workflow flagged 11 IQR outlier rows.
```

Reporting and plotting

Every workflow returns a `testflow` object. Use `report(x)`, `plot(x)`, and `as_tibble(x)`.